# SOC-in-a-Box Project: Step-by-Step Workflow and Explanation

## SETUP PHASE

### 1. Prepare the Environment

- Install a virtual machine on your Mac (e.g., using VirtualBox or UTM).
- Install a Linux OS (Ubuntu recommended) inside the VM.
- Ensure Docker, Docker Compose, and network access are working in the VM.

### 2. Deploy the ELK Stack

- Use Docker Compose to run:
  - Elasticsearch
  - Logstash
  - Kibana
- Configure Logstash to listen for incoming logs (e.g., from Filebeat and Wazuh).
- Verify that Kibana is accessible (usually on localhost:5601).

### 3. Install Wazuh Components

- Install Wazuh Manager inside the VM (or use the Wazuh Docker container).
- On your host Mac, install the Wazuh Agent for macOS.
- Configure the agent to forward logs to the Wazuh Manager.
- Ensure Wazuh forwards alerts to ELK using Logstash.

## LOG COLLECTION & MONITORING PHASE

### 4. Set Up Sysmon + Filebeat (on Windows or External Systems)

- Install Sysmon on a Windows system or VM using a security configuration file.

- Install Filebeat and configure it to:
  - Read logs from Sysmon.
  - Send logs to Logstash or directly to Elasticsearch.

5. Deploy Zeek in the VM

- Set up Zeek on the same Linux VM.
- Configure Zeek to monitor a network interface or load .pcap files.
- Ensure it logs DNS, HTTP, SSL, and connection data.
- Forward Zeek's log files to Logstash using Filebeat.

6. Install and Configure Velociraptor

- Deploy Velociraptor server in the VM.
- Install the Velociraptor client agent on endpoints (Mac, external VMs, etc.).
- Use Velociraptor to run forensic queries (e.g., scan for malicious scripts).
- Export results as logs or reports; forward them to ELK if needed.

DETECTION AND ANALYSIS PHASE

7. Centralize Everything in ELK

- Elasticsearch stores all incoming logs (from Wazuh, Filebeat, Zeek, Velociraptor).
- Logstash filters, enriches, and routes logs into proper indices.
- Kibana is used to:
  - Create dashboards (e.g., Sysmon events, Zeek connections, Wazuh alerts).
  - Visualize attacks (e.g., failed logins, suspicious network activity).
  - Set alerts or timelines.

8. Hunt for Threats

- Use Kibana's Discover and Dashboard views to search for:

  - Unusual parent-child processes from Sysmon

  - Unexpected external DNS queries from Zeek

  - Wazuh alerts for root escalation or unauthorized file access

  - Suspicious scheduled tasks detected via Velociraptor

Simulate Incidents (Optional)

9. Simulate Attacks or Threat Events

- Run Atomic Red Team tests, or simulate:

  - Phishing emails

  - Lateral movement

  - Malicious PowerShell scripts

- Observe how logs flow into your ELK stack and how alerts are triggered.

Review & Iterate

10. Fine-Tune Configurations

- Tune Filebeat paths and Logstash rules.

- Add new detection rules to Wazuh.

- Expand Velociraptor hunt packs.

- Create custom dashboards in Kibana.